

Neural network Formulas

Sigmoid Function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

ReLU Function

$$f(x) = \max(0, x)$$

Tanh Function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Softmax Function

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$